

Protective effects of a kampo medicine, Hochu-ekki-to (TJ-41) on lethal malarial infection with *Plasmodium chabaudi* AS in A/J mice

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Abstract Effects of Hochu-ekki-to (TJ-41) on the course of lethal rodent malarial infection with *Plasmodium chabaudi* AS were examined in male A/J mice. We examined the mortality, parasitemia and serum cytokines such as IL-12, IFN- γ and IL-4 in the infected and TJ-41-treated/infected mice. There was a significant difference in mortality between infected and treated/infected mice. A high mortality was observed in male mice after infection with *P. chabaudi* AS. In mice treated with TJ-41, control of the primary infection was achieved, and significantly lower mortality was observed. All surviving males in the treated/infected group showed somewhat smaller peak parasitemias than those in infected controls. Mice in the infected and treated/infected groups displayed significantly elevated serum IL-12 levels on day 4 of infection when compared with the levels from the uninfected animals. Mice in the infected and treated/infected groups displayed significantly elevated serum IFN- γ levels when compared with the levels from the uninfected animals. Furthermore, a signi-

ficantly higher IFN- γ level was seen in the treated/infected group than that in the infected group on day 4 of infection. The present results suggest that an early production of IFN- γ in the TJ-41-treated/infected mice is associated with a decrease of parasitemia, being responsible for the survival of mice.

Keywords *Plasmodium chabaudi* AS · Herbal medicine · Hochu-ekki-to · Cytokines · IFN- γ

Introduction

Traditional herbal medicines are being re-evaluated in the clinical field because of their relatively few side effects and suitability for long-term administration as compared to synthetic drugs. Recently, several kinds of these herbal medicines have been used clinically for various diseases in Japan. Numerous reports have been described on the biological actions of kampo herbal medicines on immune responses in bacterial [1] and viral infections [2], but few reports in parasitic infections. Juzen-taiho-to (TJ-48) has been reported to have immuno-potentiating activity [3] and to enhance phagocytosis [4]. The effects of TJ-48 on the course of non-lethal rodent malarial infection were examined, suggesting that it has a suppressive effect on parasitemia caused by stimulation of host immunity [5]. Regarding Hochu-ekki-to (TJ-41), several workers have reported that this herbal medicine promotes certain biological activity such as enhancement of NK activity [6], mitogenic activity [7] and macrophage activity [8].

An understanding of host–parasite interactions in malaria is an important prerequisite for chemotherapy and vaccine design. In order to gain a better insight into in vivo interactions, various murine model systems have been

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developed. Different strains of mice vary in their ability to control and clear a primary infection with *Plasmodium chabaudi chabaudi*, and immunity requires the presence of both CD4⁺ T cells and B cells [9, 10]. *Plasmodium chabaudi* AS infection in resistant mice induces sequential involvement of Th1- and Th2-regulated responses [11]. An increase in IFN- γ production is observed during the acute primary parasitemia of a *P. chabaudi* infection [12] and is thought to be essential for IL-12- and NO-dependent protection [13]. The Th2-dependent control of *P. chabaudi* infection immediately following acute primary parasitemia is thought to involve IL-4 and antibody-dependent mechanisms [14]. Although TJ-41 is able to enhance the host immune responses, few basic studies regarding its anti-parasitic effect have been reported. In this paper, we examined the antiplasmodial effect of TJ-41 in mice infected with *P. chabaudi* AS, and we also investigated the kinetics of appearance of cytokines such as IL-12, IFN- γ and IL-4 in sera.

Materials and methods

Animals and parasites

All animal experiments were performed according to the Guidelines for Animal Experimentation, Hamamatsu University School of Medicine. Inbred male and female A/J, BALB/c Cr and C57BL/6 Cr mice, 8 weeks old, were purchased from SLC Inc. (Hamamatsu, Japan) and were used as a host. The rodent malaria parasite, *P. chabaudi* AS, was maintained by serial blood passage in ICR mice, and blood-stage parasites were stored in a -80°C deep freezer.

Infection of mice with malaria parasites

For experiments, the blood-stage parasites from frozen stock were injected into two male ICR mice. The mouse showing 10–15% of parasitemia was bled under ether anesthesia to collect *P. chabaudi* AS-parasitized erythrocytes (pRBCs). Experimental mice were given an intraperitoneal injection of 10^5 pRBCs and were fed ad libitum on commercial diet, LabDiet (PMI Nutrition International, Inc., MO, USA) and water.

Compound

Hochu-ekki-to (TJ-41) was a generous gift from the Tsumura Company Limited (Tokyo, Japan) and was suspended in distilled water to prepare the stock solution at the concentration of 0.2 g/ml. The suspension was orally given to mice in a volume of 0.1 ml/10 g body weight (bw).

Experimental designs

Mouse susceptibility to *P. chabaudi* AS infection

To compare the susceptibility to malaria infection among strains and/or sexes, we first monitored their mortality in three mouse strains: A/J, BALB/c Cr and C57BL/6 Cr. Five males and five females in each strain were injected with 10^5 pRBCs intraperitoneally and were observed for days of survival up to 40 days.

Effects of Hochu-ekki-to (TJ-41) on *P. chabaudi* AS infection in A/J mice

Five male and five female A/J mice were used in infected and treated/infected groups, respectively. Two independent experiments were done. Mice in both groups were infected intraperitoneally with 10^5 pRBCs. For activity assay of TJ-41, treated mice were injected with 2 g/kg bw of TJ-41 suspension directly into the stomach through an esophageal tube once a day from 2 days before infection for 12 days. Control mice were given distilled water according to the same schedule. Following 10^5 *P. chabaudi* AS-pRBCs infection, thin blood smears were made by bleeding via the tail vein and stained with Giemsa solution throughout an observation period to determine the course of infection being expressed as % parasitemia.

Analysis of cytokines in sera during infection

We investigated the cytokines, IL-12, IFN- γ and IL-4, during the course of infection in *P. chabaudi* AS-infected A/J mice treated with TJ-41. Three normal mice were killed at the time of parasite infection. Five mice from both the infected and treated/infected groups were bled under ether anesthesia at 4 and 7 days post-infection. Three normal mice treated with TJ-41 were killed on day 10 after the first dose of the treatment regimen. Each blood collected in a tube was centrifuged, and the resultant serum was stored separately at -80°C until the day of assay. Thin blood smears were made by bleeding via the tail vein and stained with Giemsa solution at the same time of sacrifice to determine the course of infection being expressed as % parasitemia.

Cytokine assays

The levels of IL-12, IFN- γ and IL-4 were determined by sandwich ELISA [15]. For the assay, the following combinations of capture and biotinylated mAbs were used as recommended by the manufacturer (PharMingen, San Diego, CA): 9A5 and C17.8 for IL-12 (p70), R4-6A2 and XMGI.2 for IFN- γ , and 11B11 and BVD6-24G2 for IL-4. The amount of cytokines was calculated by using standard

murine recombinant cytokine curves run on the same immunoplate.

Statistical analysis

The statistical significance of differences in mortality and in serum cytokine levels between infected and treated/infected mice was determined by the Fisher's exact test and the Mann–Whitney *U* test, respectively, with $P < 0.05$ being considered significant.

Results

Course of *P. chabaudi* AS infection in A/J, BALB/c Cr and C57BL/6 Cr mice

First, we examined the mortality in male and female mice among three strains. As shown in Fig. 1, no mortality was seen in BALB/c Cr and C57BL/6 Cr mice, either male or female, following 10^5 *P. chabaudi* AS-pRBCs infection. On the other hand, A/J mice showed the susceptibility to intraperitoneal infection with the same dose of *P. chabaudi* AS-pRBCs. All of five male and two out of five female A/J mice died from 10 to 12 days post-infection (dpi). There were significant differences (Fisher's exact test $P < 0.01$) in male mortality between A/J and two other strains (BALB/c Cr and C57BL/6 Cr) against this infection.

Effects of Hochu-ekki-to (TJ-41) on *P. chabaudi* AS infection in A/J mice

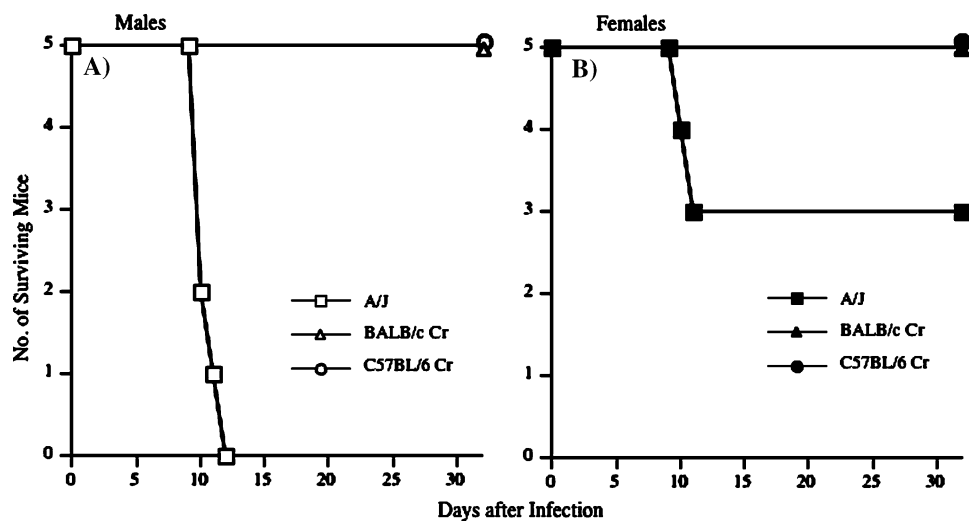
Mice were orally given 2 g/kg bw of TJ-41 suspension once a day from 2 days before infection for 12 days. The data for mouse survival have been pooled from two com-

parable experiments. TJ-41 showed curative effects against 10^5 *P. chabaudi* AS-pRBCs infection in male A/J mice (Fig. 2a). Eight of 10 male A/J mice in the infected control group died between 9 and 11 dpi. In contrast, 2 of 10 males died in the TJ-41-treated group. There was a significant difference ($P < 0.05$, Fisher's exact test) in male mortality between infected and treated/infected mice. On the other hand, TJ-41 showed no curative effects in female mice (Fig. 2b). Two and three out of ten female A/J mice in infected and treated/infected groups, respectively, died by 10 dpi. Surviving mice in infected and treated/infected groups, both males and females, showed primary and small secondary peaks of parasitemias at 7–9 dpi and 17–19 dpi, respectively (Fig. 3). In the infected group, two males who had recovered from this infection showed a primary peak parasitemia of 50.9 and 52.1%, respectively. These values were somewhat lower than those ranging between 55.1 and 62.3% observed in surviving female mice. Other dead male mice showed primary peak parasitemias of more than 67% before death. Three dead female mice showed higher peak parasitemias (67.1, 68.5 and 70.7%) than those of surviving female mice. In treated/infected groups, all surviving males showed primary peak parasitemias of less than 62%. There is no difference in primary peak parasitemia between infected female control and treated/infected groups.

Analysis of cytokines in sera during the course of infection

The kinetics of appearance of cytokines such as IL-12, IFN- γ and IL-4 in the sera from both males and females was examined (Figs. 4, 5, 6). The mean value of serum IL-12 in the normal male mice was 31.0 ± 2.1 pg/ml (as mean $\pm$ SD, $n = 3$) (Fig. 4a). The mean value of serum IL-12 in the TJ-41-given mice was 26.7 ± 3.6 pg/ml ($n = 3$)

Fig. 1 The outcome of infection with *P. chabaudi* AS in A/J, BALB/c Cr and C57BL/6 Cr. Male **a** and female **b** mice in each strain, 8 weeks old, were infected intraperitoneally with 10^5 pRBCs. Survival of mice was monitored daily following infection until the termination of the experiment. There was a significant difference (Fisher's exact test $P < 0.01$) in male mortality between A/J and two other strains (BALB/c Cr and C57BL/6Cr) against 10^5 *P. chabaudi* AS-pRBCs infection



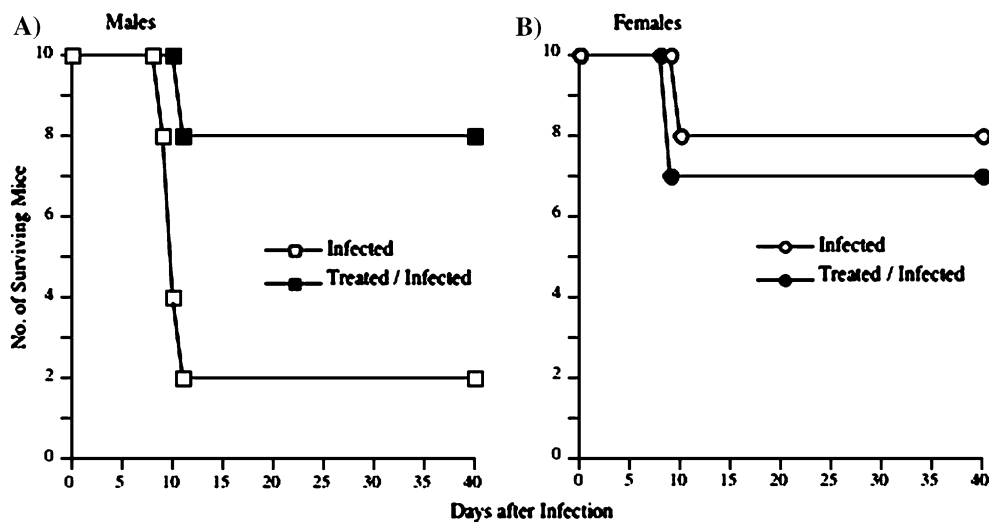
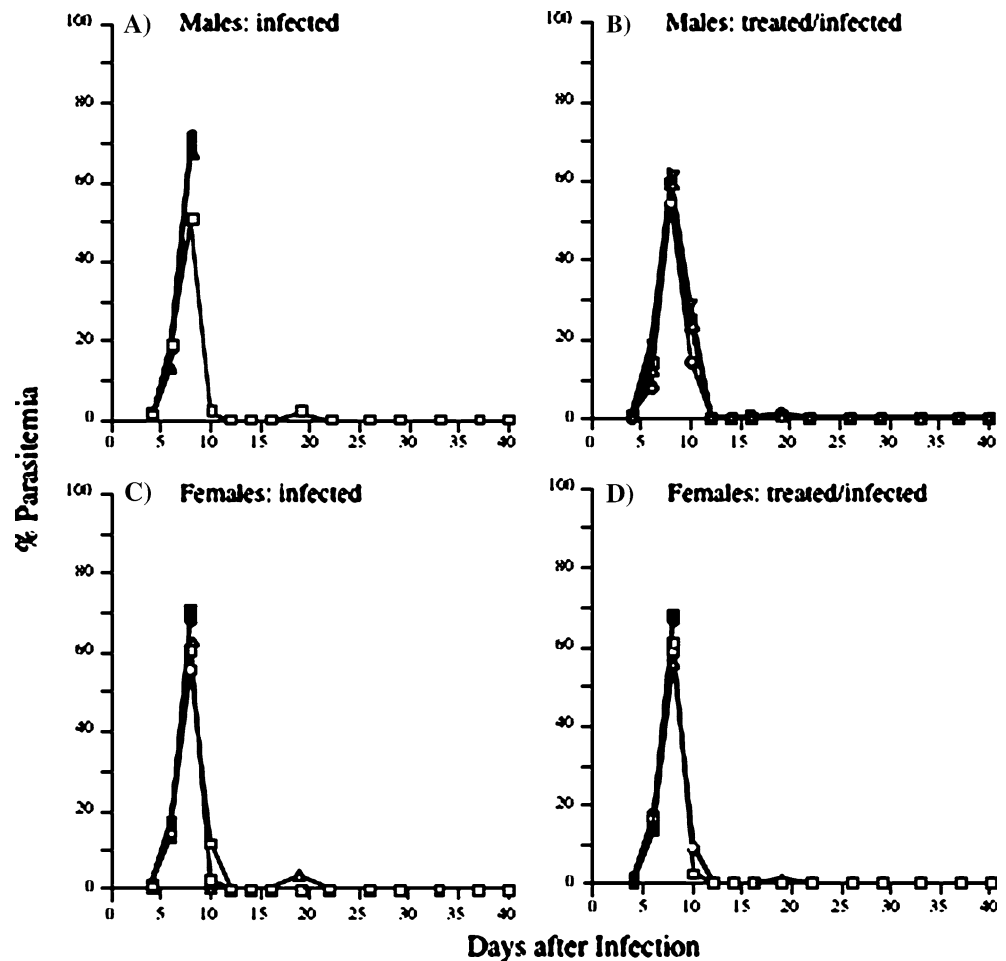


Fig. 2 The outcome of infection with *P. chabaudi* AS in A/J mice in response to treatment with Hochu-ekki-to (TJ-41). **a** Male and female A/J mice, 8 weeks old, were given 2 g/kg bw of TJ-41 suspension once a day from 2 days before intraperitoneal injection with 10^5 pRBCs for 12 days. Infected control mice were given distilled water

according to the same schedule. Survival of mice was monitored daily following infection until the termination of the experiment. The data shown are representative of two comparable experiments with five mice. There was a significant difference (Fisher's exact test $P < 0.05$) in male mortality between infected and treated/infected mice

Fig. 3 Parasitemia profile in the bloodstream of A/J mice in response to treatment with Hochu-ekki-to (TJ-41). Male and female A/J mice, 8 weeks old, were given 2 g/kg bw of TJ-41 suspension once a day from 2 days before intraperitoneal injection with 10^5 *P. chabaudi* AS-parasitized RBCs for 12 days. Infected control mice were given distilled water according to the same schedule. Each symbol represents an individual mouse (white symbols for survival and black ones for dead). **a** Male infected (survival $n = 1$; dead $n = 4$), **b** male treated/infected (survival $n = 5$), **c** female infected (survival $n = 3$; dead $n = 2$), and **d** female treated/infected (survival $n = 3$; dead $n = 2$)



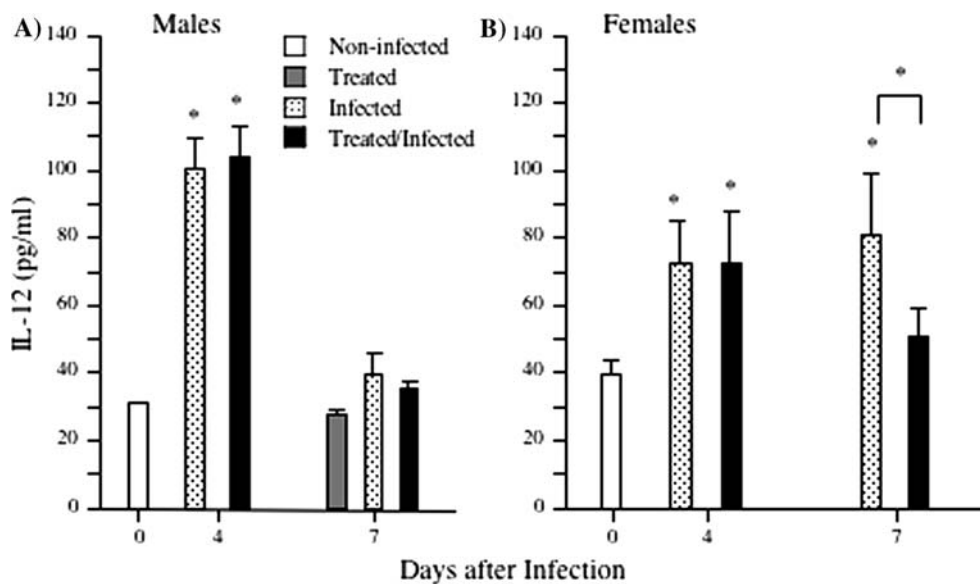


Fig. 4 Course of serum IL-12 levels determined by ELISA in male **a** and female **b** A/J mice infected with 10^5 *P. chabaudi* AS. Ten treated mice, 8 weeks old, were given 2 g/kg bw of Hochu-ekki-to (TJ-41) suspension once a day from 2 days before intraperitoneal injection with 10^5 pRBCs for 12 days. Ten infected control mice were given distilled water according to the same schedule. Three normal mice

were used as uninfected controls (day 0). Following injection of the parasites, five mice in each group were sacrificed under ether anesthesia at day 4 and 7 post-infection. Three male mice treated with TJ-41, but not infected were killed at day 4. Data represent the mean values with the standard error. *Significantly different from the value of the uninfected control mice (Mann–Whitney *U* test, $P < 0.05$)

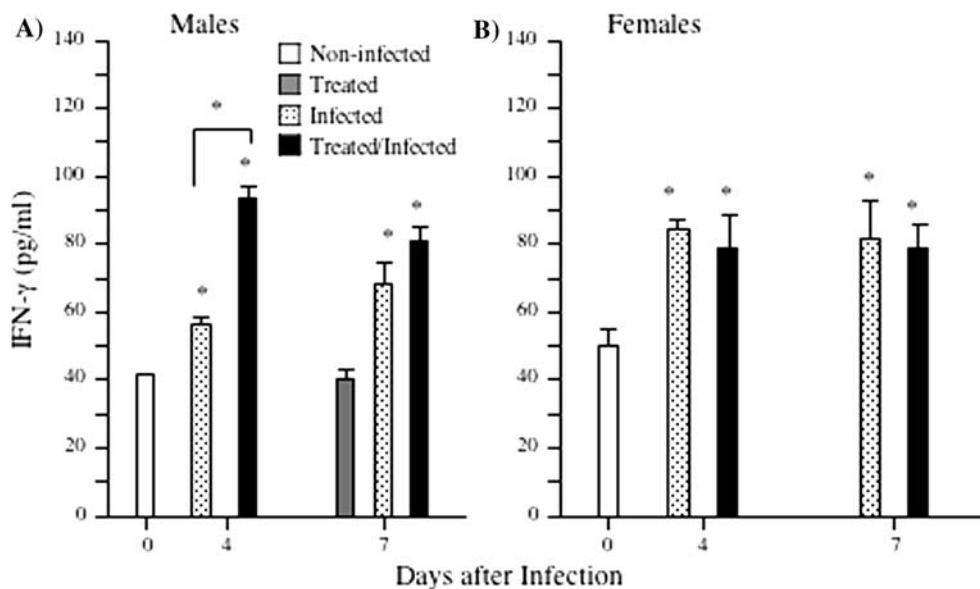


Fig. 5 Course of serum IFN- γ levels determined by ELISA in male **a** and female **b** A/J mice infected with 10^5 *P. chabaudi* AS. Ten treated mice, 8 weeks old, were given 2 g/kg bw of Hochu-ekki-to (TJ-41) suspension once a day from 2 days before intraperitoneal injection with 10^5 pRBCs for 12 days. Ten infected control mice were given distilled water according to the same schedule. Three normal mice

were used as uninfected controls (day 0). Following injection of the parasites, five mice in each group were sacrificed under ether anesthesia at day 4 and 7 post-infection. Three male mice treated with TJ-41, but not infected were killed at day 4. Data represent the mean values with the standard error. *Significantly different from the value of the uninfected control mice (Mann–Whitney *U* test, $P < 0.05$)

on day 10 after the first dose of the treatment regimen (on day 7 of infection). Male mice in the infected ($n = 5$) and the treated/infected ($n = 5$) groups displayed significantly

elevated serum IL-12 levels on 4 dpi when compared with the levels from the normal mice ($P < 0.05$ Mann–Whitney *U* test). There was no significant difference in IL-12 level

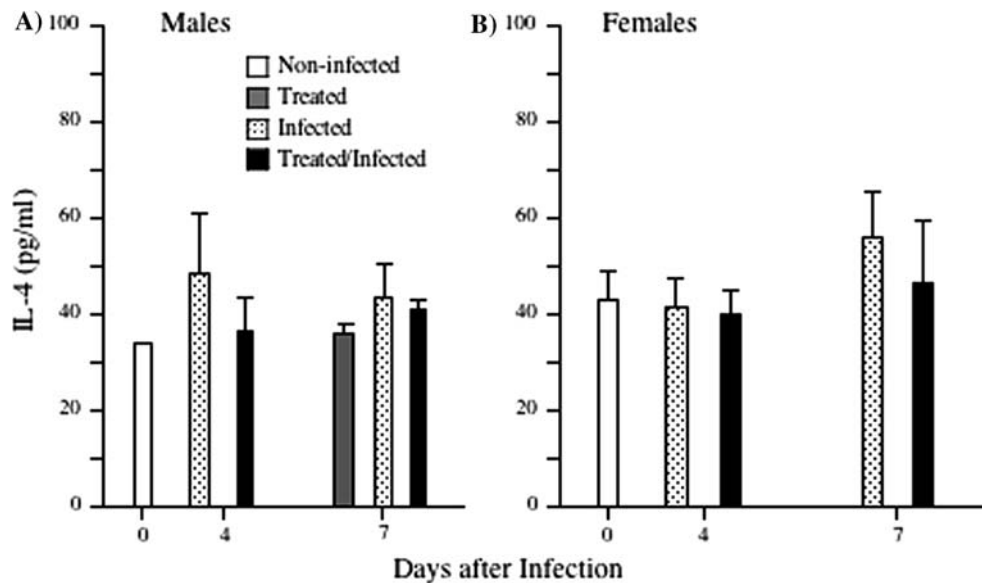


Fig. 6 Course of serum IL-4 levels determined by ELISA in male A/J mice infected with 10^5 *P. chabaudi* AS. Ten treated mice, 8 weeks old, were given 2 g/kg bw of Hochu-ekki-to (TJ-41) suspension once a day from 2 days before intraperitoneal injection with 10^5 pRBCs for 12 days. Ten infected control mice were given distilled water according to the same schedule. Three normal mice were used as

uninfected controls (day 0). Following injection of the parasites, five mice in each group were sacrificed under ether anesthesia at day 4 and 7 post-infection. Three male mice treated with TJ-41, but not infected were killed at day 4. Data represent the mean values with the standard error

between the normal and other groups. The mean value of serum IFN- γ in the normal mice was 41.4 ± 2.9 pg/ml ($n = 3$) (Fig. 5a). The mean value of serum IFN- γ in the TJ-41-given mice was 39.4 ± 6.3 pg/ml ($n = 3$). Mice in the groups except for the TJ-41-given group displayed significantly elevated serum IFN- γ levels when compared with the levels from the normal mice ($P < 0.05$). Furthermore, there was a significant difference in IFN- γ level between the infected and treated/infected groups on 4 dpi ($P < 0.05$). The mean value of serum IL-4 in the normal mice was 32.1 ± 1.0 pg/ml ($n = 3$) (Fig. 6a). The mean value of serum IL-4 in the TJ-41-given mice was 35.1 ± 5.6 pg/ml ($n = 3$). There was no significant difference in the IL-4 level between the normal and other groups though mice in the infected group showed some increase of this cytokine at 4 dpi.

In the normal female mice, the mean value of serum IL-12 was 38.7 ± 8.9 pg/ml (as mean $\pm$ SD, $n = 3$) (Fig. 4b). Female mice in the infected ($n = 5$) and the treated/infected ($n = 5$) groups displayed significantly elevated serum IL-12 levels on 4 dpi when compared with the levels from the normal mice ($P < 0.05$ Mann–Whitney *U* test). A significant elevation of IL-12 values was also seen on 7 dpi in the infected mice ($P < 0.05$). Furthermore, there was a significant difference in this cytokine level between in the infected and the treated/infected groups on 7 dpi ($P < 0.05$). The mean value of serum IFN- γ in the normal mice was 49.5 ± 11.2 pg/ml ($n = 3$) (Fig. 5b). Mice in the

infected and treated/infected groups displayed significantly elevated serum IFN- γ levels when compared with the levels from the normal mice on 4 and 7 dpi ($P < 0.05$). The mean value of serum IL-4 in the normal mice was 42.3 ± 11.6 pg/ml ($n = 3$) (Fig. 6c). There was no significant difference in IL-4 level between the normal and other groups, though mice in the infected group showed some increase of this cytokine at 7 dpi.

Discussion

Differences of mouse strain and/or sex in susceptibility to *P. chabaudi* AS infection allow us to study immunological components involved in malaria pathology. In accordance with previous reports [16, 17], the present results showed the high mortality of A/J mice as compared with BALB/c Cr and C57BL/6 Cr mice following *P. chabaudi* AS infection. Protective immunity against the infection with *P. chabaudi* was shown to be mediated by the activation of Th1 cells, and IFN- γ was demonstrated to play a critical role in the control of malarial infection [12, 18]. Early appearance of IFN- γ is associated with resistance to infection [19]. Clark et al. [20] demonstrated inhibition of *P. chabaudi* infection by injection of recombinant IFN- γ in CBA mouse strain. Stevenson et al. [18] described the role of endogenous IFN- γ in host response to *P. chabaudi* infection in C57BL/6 mice. In the present study, the bio-

logical effect of Hochu-ekki-to (TJ-41) on host immunity in *P. chabaudi* infection was investigated in susceptible A/J mice. By the analysis of cytokines in sera, TJ-41-given, infected mice showed a significantly higher level of IFN- γ than that in infected mice at day 4 of infection, suggesting the involvement of herbal medicine in the higher production of this cytokine. This finding suggests that the enhancement of IFN- γ -related mechanisms is associated with the survival of treated mice following infection.

While some pathology occurs only with certain combinations of parasite and inbred mouse strains, anemia is a feature of all rodent malaria infections [21]. Though both resistant C57BL/6 and susceptible A/J mice showed anemia after intraperitoneal infection with *P. chabaudi* AS, A/J mice developed severe anemia and succumbed by days 8–12 post-infection, but C57BL/6 mice experienced a less severe level of anemia and eventually recovered from the infection [22]. Importantly, malarial anemia coincides with a marked increase in parasitized erythrocytes. The present results showed a decrease in peak parasitemia in the surviving TJ-41-treated, infected male A/J mice relative to the infected controls, and hence an early production of IFN- γ is associated with a decrease of parasitemia being responsible for the survival of mice. On the other hand, hypothermia and hypoglycemia were more marked in susceptible mice than in resistant animals, and hypothermia particularly correlated closely with parasite levels [17]. Further studies on lethality in *P. chabaudi* AS infections may clarify the effect of herbal medicines on survival in malarial infections.

The high mortality of *P. chabaudi* AS-infected male A/J mice was abolished by treatment with herbal medicine, but no improvement was seen in female mice. This sex difference is an interesting finding from the viewpoint of a possible involvement of sex hormones in protection against malarial infection. The male hormone, testosterone, has been shown to inhibit the host ability to overcome *P. chabaudi* infection [23, 24]. Recently, using the genetic-engineered mutant mice, Zhang et al. [25] suggested that the male sex hormones modulate the function of Th1/Th2 cells in the host response to *P. chabaudi* AS infection, and the in vivo mechanisms involved in protection against this parasite infection are more complex than a simple Th1/Th2 dichotomy. In addition, since neither orchidectomy nor ovariectomy affected the parasitemia of either IFN- γ R or IL-4 knockout mice, they also suggested that different mechanisms operate in preventing death from the infection and inhibiting the parasitemia. No improvement in mortality of female mice was seen by treatment with TJ-41, and thus a detailed analysis of counterbalancing between the immune and the endocrine systems may be key to defining the effect of TJ-41 on *P. chabaudi* AS infection in A/J mice.

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